

Hidden markov mark-recapture modeling highlights transboundary exchange for northeast pacific sablefish

Maia Kapur^{a,*}, Luke Rogers^{1,b}, Daniel R. Goethel^c, Matthew L.H. Cheng^c, Dana Hanselman^c, Katy Echave^c, Melissa A. Haltuch^a, Kendra R. Holt^b, Philina English^b, Aaron M. Berger^d, Brendan Connors^e

^a NOAA Fisheries, Alaska Fisheries Science Center, Seattle, WA, USA

^b Pacific Biological Station, Fisheries and Oceans Canada, 3190 Hammond Bay Road, Nanaimo, British Columbia V9T 6N7, Canada

^c NOAA Fisheries, Alaska Fisheries Science Center, Juneau, AK, USA

^d NOAA Fisheries, Northwest Fisheries Science Center, Newport, OR, USA

^e Institute of Ocean Sciences, Fisheries and Oceans Canada, 9860 W Saanich Rd, Sidney, British Columbia V8L 4B2, Canada

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ABSTRACT

Developing robust fisheries management advice requires understanding fish population distribution and movement in relation to harvesting and management boundaries. Traditional stock assessment models often assume homogeneous populations with negligible transboundary movement, an assumption that may lead to undesirable management outcomes when violated. We developed a hidden Markov movement model and fitted it to almost forty years (1979–2018) of conventional tagging data for northeast Pacific sablefish (*Anoplopoma fimbria*), a commercially valuable species distributed across the northeast Pacific. This is the first time that sablefish movement has been evaluated at this spatial scale and included transboundary migration. Estimated movement rates were then combined with stock assessment abundance estimates to quantify the magnitude of exchange between management regions. Several sensitivity analyses were conducted to evaluate the robustness of the model assumptions regarding size structure, time-at-liberty, size-transition and parameter uncertainty. Sablefish exhibited long-distance movements. Notably, British Columbia acted as a transition zone for sablefish movement between regions and a net recipient for sablefish abundance, receiving net immigration from both Alaska and the California Current, a pattern influenced by regional abundance differences. Differences in movement rates between size classes were modest, although larger sablefish in Alaska were more likely to move into British Columbia compared to smaller fish. Our findings, consistent with previous smaller-scale sablefish movement research, uniquely quantify the demographic exchange across management regions, highlighting a large contribution of sablefish from Alaska and the California Current to British Columbia. This study emphasizes the need to account for the transboundary nature of sablefish within assessment and management frameworks. Collaborative, international efforts in monitoring and data sharing will be essential for ensuring the sustainable harvest of transboundary stocks like sablefish.

1. Introduction

Developing robust management advice relies on understanding the spatiotemporal distribution of fish populations in relation to both harvesting as well as assessment and management boundaries. An underlying tenet of most stock assessment models that are used to inform

fisheries management decision-making and catch quota determination is that a well-mixed, homogeneous unit population exists in the area being assessed and managed (Beverton and Holt, 1957). Moreover, immigration or emigration across primary assessment boundaries or among adjacent management units is often assumed to be negligible or non-existent (Goethel et al., 2023a). However, because boundaries are

* Corresponding author.

E-mail address: kapur.mr@gmail.com (M. Kapur).

¹ Maia Kapur and Luke Rogers shared first authorship on this article.

² Present address: Pacific Northwest National Laboratory, 902 Battelle Boulevard, Richland, WA 99352

often human constructs (Song et al., 2017), violations of assessment assumptions inherently occur. Unfortunately, management and assessment units have historically been defined based on harvest trends and political convenience in many cases with limited consideration of underlying biological processes (Berger et al., 2017; Kerr et al., 2017). Transboundary, straddling (i.e., those that transit both the open ocean and various countries' exclusive economic zones, EEZs), and highly migratory stocks further exacerbate the ability to define adequate management boundaries (Cadurin et al., 2023).

Population structure within a species may result in multiple population components within a single assessment or management unit, a single population component spanning multiple assessment or management units, or, ideally, each assessment or management unit encompassing a single population component (Berger et al., 2021, 2017). When underlying population structure is not accounted for in the delineation of assessment and management boundaries, it often results in biased assessment outputs with over- or under-utilization of various population components (Kapur et al., 2024; Kerr et al., 2017). It is widely acknowledged that many well-publicized fishery collapses can be linked to failure to account for spatial population structure and connectivity dynamics among population components (e.g., northern cod; Smedbol and Stephenson, 2001; Petitgas et al., 2010; Ciannelli et al., 2013). However, imperfect boundaries can still enable adequate management if they encompass the majority of the population biomass distribution (Berger et al., 2021). With improved recognition of the spatial complexity in marine fish populations, the adequacy of existing boundaries and assessment approaches need to be periodically reassessed as new data are collected. Similarly, the increasing implementation of spatial management measures (e.g., marine protected areas, MPAs) necessitates improved high resolution data sets that can inform spatiotemporal distributions and interactions across management boundaries (Berger et al., 2017).

Ecosystem variability is leading to new movement patterns and redistribution of numerous marine species, which has caused management complications as fish increasingly move among and into new management units (Gullestad et al., 2020; Liu et al., 2023; Mendenhall et al., 2020; Vogel et al., 2023). Furthermore, the ability to adequately monitor and assess species is also confounded by the movement of fish outside of traditional survey footprints as ecosystems rapidly change. For instance, inappropriate catch recommendations may result if changes in the availability of a species to a survey are not detected or addressed (e.g., O’Leary et al., 2022, 2021). Thus, identifying movement and migration corridors and changes in these patterns seasonally and over time will be inherently necessary to ensure sustainable future management and to inform marine spatial planning decisions (Gaines et al., 2018; Lowerre-Barbieri et al., 2019). Fisheries management must similarly determine the degree to which existing boundaries adequately delineate an exploited resource, by quantifying movement across them (Cadurin et al., 2023; Lédée et al., 2021).

In the face of the challenges posed by imperfect understanding of population structure, mismatch with management, and potential for changes in ecosystem processes to exacerbate them, tagging methods that can help elucidate and monitor rangewide connectivity are critical. Although a myriad of tag types can inform movement and migration pathways, a major impediment towards understanding rangewide distributions is the scaling from individual tagged animals to population-level movement patterns (Jeltsch *et al.*, 2013; Plard *et al.*, 2019). Conventional tags are inexpensive and can be deployed with large sample sizes over long experiments potentially spanning decades and multiple ecological regimes if a consistent experimental survey exists (Hanselman *et al.*, 2014). However, data from each conventional mark-recapture event are limited, given that they only provide release and recapture positions, which make inference of migration paths difficult. There have been recent developments in methods to translate individually-implanted electronic archival and satellite pop-up tags into full migration paths (e.g., Griffiths *et al.*, 2018). However, these tag

types suffer from cost and sample size limitations, and often have limited time series compared to conventional tags (Crossin et al., 2017; Sippel et al., 2015). Thus, each tag type can provide important insight into population structure and movement, but conventional tagging data sets with long time series still provide important insight into broadscale spatiotemporal trends and distributions.

Sablefish (*Anoplopoma fimbria*) are a long-lived, commercially valuable fish that is widely distributed across the northeastern Pacific Ocean, with accessible data spanning three EEZs (United States, Canada, and Mexico; [Rooper et al., 2021](#)). Sablefish are known to exhibit long range movements (e.g., 1000 s of kilometers) in short periods of time (e.g., less than a year; [Hanselman et al., 2014](#); [Kimura and Shimada, 1998](#)), thereby, potentially interacting with different fisheries and fishing intensities across their lifespan. A single panmictic population is thought to exist across their range based on genetics ([Jasonowicz et al., 2016](#), [Timm et al., 2024](#)), but some differentiation may be present at range extremes (e.g., in Mexico and Russia; [Orozco-Ruiz et al., 2023](#)); these size of these regional populations are so small as to be irrelevant as they exist at the tail end of the species' range. The impact of environmental changes on sablefish dynamics and distributions is not fully understood, but warmer ocean temperatures have been associated with improved recruitment success and population expansion into historical population strongholds (e.g., the eastern Bering Sea Shelf in Alaska; [Cheng et al., 2025](#)). Similarly, movement rates of juvenile sablefish in Alaska appear linked to year-class strength with indications that movement rates have potentially increased in the 2010s compared to historical estimates ([Hanselman et al., 2014](#)). This coincides with a change from generally low recruitment in the early 2000s, leading to sablefish in Alaska approaching lower biomass target levels for management, followed by high recruitment events in the mid-2010s which increased biomass above levels associated with precautionary harvest advice ([Goethel et al., 2022](#)). Given the complexity of sablefish management (i.e., multiple regional and country jurisdictions working in isolation) along with the ability of sablefish to readily move across management boundaries, it is necessary to develop a thorough understanding of the amount of exchange and degree of interannual variability in movement across boundaries.

Sablefish movement dynamics in the northeast Pacific have been widely studied since the 1970s with long-term tagging studies that have been implemented by various fisheries agencies (e.g., National Oceanic and Atmospheric Administration, Alaska Department of Fish and Game, Japan Fisheries Agency, and Department of Fisheries and Oceans Canada). Past work has focused on movement within, and out of, specific jurisdictions like Alaska (Hanselman et al., 2014; Heifetz and Fujioka, 1991; Maloney and Sigler, 2008; Cheng et al. In Review), British Columbia (Beamish and McFarlane, 1988), and the U.S. West Coast in the California Current (Kimura and Shimada, 1998). These studies have found that in general sablefish exhibit high movement rates that vary by region (e.g., Hanselman et al., 2014; Heifetz and Fujioka, 1991), in relation to oceanographic conditions (Kimura and Shimada, 1998), and spawning biomass (Hanselman et al., 2014), while also demonstrating strong ontogenetic patterns (Heifetz and Fujioka, 1991; Cheng et al., 2025). Some tagging studies have also suggested the potential for sexually dimorphic movement patterns, with females exhibiting greater movement potential than males (Morita et al., 2012). There appears to be some disagreement among studies regarding the presence of directional movement patterns, with some studies supporting this phenomenon and others finding little evidence (Hanselman et al., 2014; Heifetz and Fujioka, 1991; Cheng et al., 2025). Despite the breadth of research on sablefish movement, previous tagging analyses were limited in spatial extent and unable to estimate movement rates across the entire northeastern Pacific sablefish distribution, with the exception of Kimura et al. that used federal tags only to descriptively identify movement patterns. In addition, previous work has not paired movement rates with regional population trends to inform relative and absolute exchange of numerical abundance among regions. To help fill these knowledge gaps,

we quantify sablefish movement rates over time across the entire northeast Pacific using state and federal tagging data for the first time, and then translate the resulting movement rates into consequences for harvest management decisions (i.e., changes in abundance among international management jurisdictions).

2. Methods

We developed a hidden Markov movement model (HMM) and fit it to almost forty years of tagging data to quantify sablefish movement rates by size class and among management regions in the northeast Pacific. The HMM is a preferred tool for quantifying movement from tag-recapture data that only consists of discrete release and recovery locations with time stamps because the status and location of individual fish between tagging and recapture events are unobserved (hidden). This framework accounted for the probabilistic nature of fish movement conditional on state-dependent characteristics (such as size). The particular strength of this approach allowed us to account for observational uncertainty via region-specific tag reporting rates as well as modeling fishing mortality as a separate process, all of which contributed to the likelihood of tag recovery. This allowed our analysis to appropriately quantify statistical uncertainty driven by two key processes that vary across management jurisdictions (fish size and exploitation levels) and facilitate sablefish movement insights across an unprecedented spatial scale.

We then paired the estimated movement rates with estimates of sablefish abundance from recent stock assessments to quantify annual abundance exchange among Alaska, British Columbia, and the

California Current. A detailed description of the data, models, model fitting and inference is provided below. All data and code to reproduce the analyses are archived at <https://github.com/luke-a-rogers/sablefish-movement-2024>. A suite of sensitivity analyses enabled the estimation of movement rates for immature and mature size classes separately, and to evaluate the impact of key model assumptions on our findings.

2.1. Data

We used sablefish tag release and recovery observations from conventional tagging conducted in the northeast Pacific from 1979–2018 (Fig. 1). Tagged sablefish were released from National Marine Fisheries Service (NMFS) longline surveys along the continental slopes of Alaska and the California Current (Sigler, 2000), by Alaska Fish and Game (ADF&G) within coastal waters of Southeast Alaska (Carlile et al., 2002), and from Fisheries and Oceans Canada (DFO) longline trap surveys along the continental slope and in four mainland inlets of British Columbia (Lacko et al., 2020; Wyeth et al., 2005) (Fig. 1). Tags from NOAA, DFO, and ADF&G were released almost continuously during this period; tagged sablefish were also released off the state of Oregon for a research collaboration with commercial fishermen during 1996–2004 (Figure S1; described in Sogard and Berkeley, 2017). Sablefish tags were recovered on fisheries independent surveys and in commercial fishery catch. The model considered three international management regions (Alaska, British Columbia and the California Current, which includes the states of Washington, Oregon and California). Supplementary analyses were performed using six further divisions within each region based on demographic (Kapur et al., 2020) and marine ecosystem breaks. Results

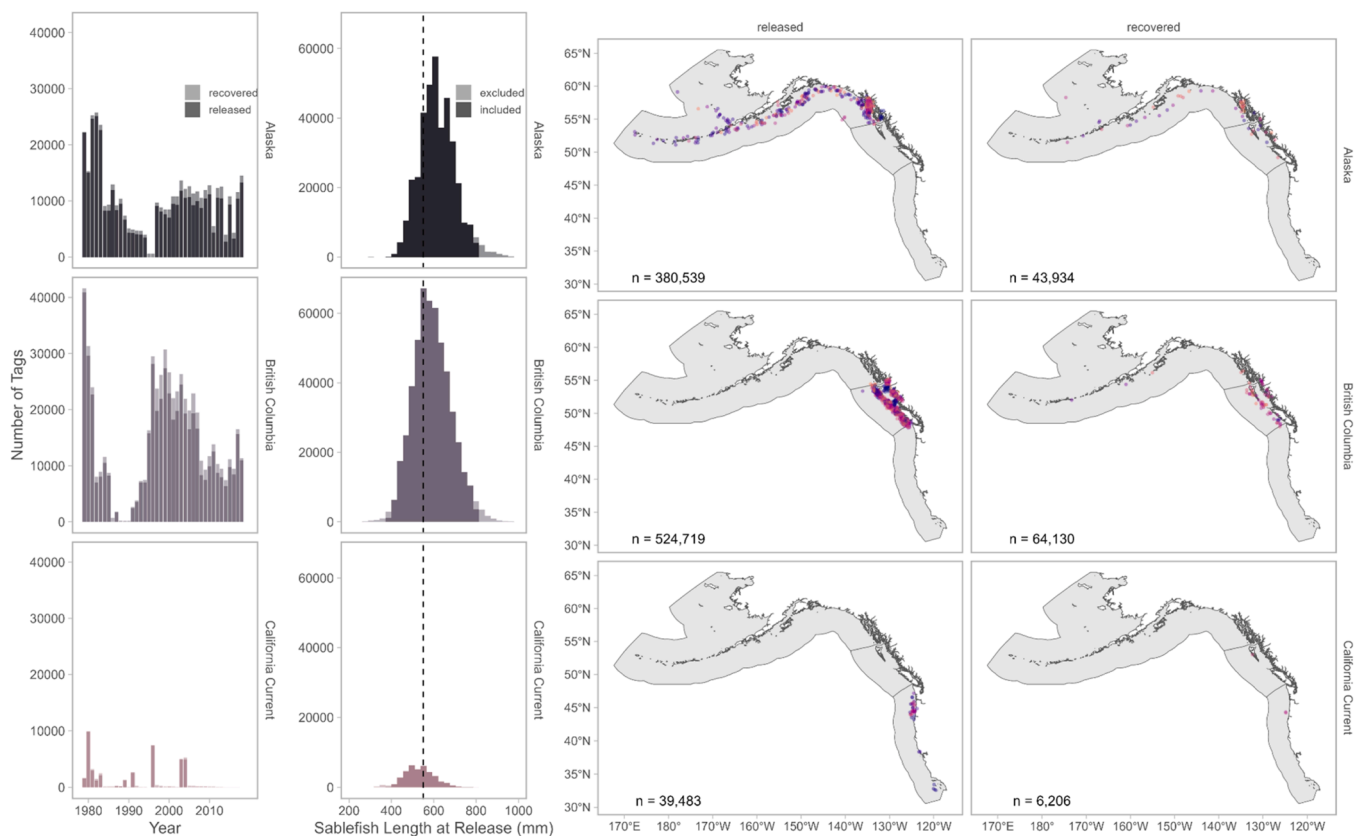


Fig. 1. Leftmost column: histogram of sablefish tag releases (opaque bars) and recoveries (transparent bars) by year and management region (rows, colors) Center column: histogram of sablefish fork length at release in mm. Transparent bars indicate excluded tags, while opaque bars are those included in this study (400–800 mm). The vertical dashed line indicates the separation between small and large size classes, 550 mm. Rightmost panel: Tagged sablefish released (third column) and recovered (fourth column) by management region (rows) during 1979–2018. Colors correspond to years, with warmer colors more recent. Points represent individual fish. Text labels indicate the number of tags in each category. Geolocations have been jittered to preserve confidentiality and therefore may appear slightly outside of political boundaries.

for the latter are not discussed here but provided in [Supplementary Material](#).

We converted the tagging data to quarterly release and recovery counts by region with a single, pooled size class (we explored movement rate estimates for immature and mature size classes as a sensitivity, described below). An important assumption underlying tagging models is that tagged fish are well-mixed across the population such that tagged and untagged fish are equally vulnerable to recapture. We therefore removed any recoveries that occurred within four months of release to ensure tagged fish had adequate time to mix. We also limited the tag release-recovery observations used in fitting the model to have a maximum time at liberty of three years (twelve quarterly model time steps). This was specified to limit size-class smearing (i.e., confounding of movement rates for a given size class with growth from one size class to the next) on our inference when the model represented more than one size class. The aforementioned specifications describe the main model, hereafter referred to as the “base” model. We explored the influence of these assumptions through a range of sensitivity analyses.

2.2. Movement model

We modeled sablefish movement as a hidden Markov process (Langrock et al., 2012) observed by tag recoveries. Movement model variables and parameter descriptions are provided in [Table 1](#) and [Table 2](#) respectively. The model used counts of sablefish tag releases $\mathbf{R}$ by release region and size class (l). It is possible to use a single size class. The initial tag counts $\mathbf{A}$ of sablefish were represented by a square diagonal matrix:

$$\text{diag}[\mathbf{A}_{i,d=1,l}] = \mathbf{R}_{i,l}(\mathbf{1} - \mathbf{v}), \quad \text{Eq. 1}$$

Where rows corresponded to release region and diagonal elements were counts of tags released in each region (i) diminished by an initial tag loss rate ν implemented during the first time step ($d=1$). There were twelve quarterly time steps modeled for each sablefish, corresponding to three years at liberty following release; in the following equations, n indexes year while d indexes sub-annual time steps. Subsequent tag counts from the same release time-step and size class were defined recursively as:

$$\mathbf{A}_{i,d,l} = \mathbf{A}_{i,d-1,l} \mathbf{\Gamma}_{n[i,d-1],l}, \quad \text{Eq. 2}$$

where $\mathbf{\Gamma}_{n+d-1,l}$ acted as a transition matrix, and $\mathbf{A}_{i,d,l}$ were the projected tag counts at the beginning of model time-step $n + d - 1$ with rows indicating the release region and columns indicating the destination region.

The transition matrix $\mathbf{\Gamma}_{n,l}$ is the matrix product of the survival and movement rates:

$$\mathbf{\Gamma}_{n,l} = \mathbf{S}_{t[n],l} \mathbf{\Delta}_l, \quad \text{Eq. 3}$$

where $\mathbf{S}_t$ is a diagonal matrix of stepwise survival rates in year t , $\mathbf{\Delta}_l$ is a square matrix of stepwise movement rates, and $t[n]$ indexed year as a function of model time step. While $\mathbf{\Delta}_l$ is a proper right stochastic (Markov) matrix (its elements were probabilities and each row sums to one), $\mathbf{\Gamma}_{n,l}$ is not, because mortality removed some tagged sablefish from the model during each time-step. Mortality, including tag loss, was modeled as an absorbing Markov state: once individuals entered this state, they could not transition to any other. This absorbing state existed alongside a set of non-absorbing states representing spatial regions, the number of which we varied across model configurations. For clarity, we encoded the mortality state implicitly via removals, rather than assign an additional row and column to $\mathbf{\Gamma}_{n,l}$, and we refer to $\mathbf{\Gamma}_{n,l}$ as a transition matrix for convenience with the understanding that its rows did not sum to one.

Sablefish survival rates depended on mortality and tag loss as:

$$\text{diag}[\mathbf{S}_{n,l}] = \exp[-\lambda \mathbf{f}_{t[n]} - \mathbf{m} - \eta], \quad \text{Eq. 4}$$

where $\mathbf{f}_t$ and $\mathbf{m}$ were row vectors of time-step-wise fishing and natural

Table 1

Symbols used in movement model and throughout this manuscript.

Symbol	Condition	Dimension	Definition	Type
<i>Indices</i>				
X	$\in \mathbb{Z}^+$	Scalar	Number of geographic regions	Index limit
T	$\in \mathbb{Z}^+$	Scalar	Number of years	Index limit
K	$\in \mathbb{Z}^+$	Scalar	Number of time steps per year	Index limit
N	$=TK$	Scalar	Number of time steps	Index limit
D	$\in \mathbb{Z}^+$	Scalar	Maximum time steps at liberty	Index limit
L	$\in \mathbb{Z}^+$	Scalar	Number of release size classes	Index limit
<i>Sub-indices</i>				
t	$\in [1..T]$	Scalar	Year	Index
n	$\in [1..N]$	Scalar	Model time step	Index
i	$\in [1..N - 1]$	Scalar	Release time step	Index
k	$\in [1..K]$	Scalar	Time step within year	Index
d	$\in [1..D]$	Scalar	Time step at liberty	Index
l	$\in [1..L]$	Scalar	Release size class	Index
<i>Data provided to model</i>				
R	$\in \mathbb{Z}^{+,0}$	$[N-1,L][X]$	Sablefish tags released (counts)	Stepwise
Q	$\in \mathbb{Z}^{+,0}$	$[N-1,D,L][X,X]$	Sablefish tags recovered (counts)	Stepwise
<i>Derived Quantities and Parameters</i>				
A	$\in \mathbb{R}^{+,0}$	$[N-1,D,L][X,X]$	Sablefish tag abundance (numbers)	Stepwise
$\hat{Q}$	$\in \mathbb{R}^{+,0}$	$[N-1,D,L][X,X]$	Sablefish expected tags recovered (numbers)	Stepwise
$\mathbf{r}$	$\in [0,1]$	$[N,L][X,X]$	Sablefish transition rates	Stepwise
S	$\in [0,1]$	$[N,L][X,X]$	Sablefish Survival Rates	Stepwise
B	$\in [0,1]$	$[N,L][X]$	Sablefish tag observation rates	Stepwise
P	$\in [0,1]$	$[L][X,X]$	Sablefish movement rates	Annual
Δ	$\in [0,1]$	$[L][X,X]$	Sablefish movement rates	Stepwise
F	$\in \mathbb{R}^{+,0}$	$[T,X]$	Sablefish instantaneous fishing mortality rates	Annual
f	$\in \mathbb{R}^{+,0}$	$[T,X]$	Sablefish instantaneous fishing mortality rates	Stepwise
M	$\in \mathbb{R}^{+,0}$	$[T,X]$	Sablefish instantaneous natural mortality rates	Annual
m	$\in \mathbb{R}^{+,0}$	$[X]$	Sablefish instantaneous natural mortality rates	Stepwise
λ	$\in [0,1]$	$[L,X]$	Sablefish fisheries size selectivity	Per fish
θ	$\in [0,1]$	$[X]$	Sablefish tag reporting rates	Per tag
η	$\in \mathbb{R}^{+,0}$	Scalar	Sablefish instantaneous ongoing tag loss rate	Stepwise
v	$\in [0,1]$	Scalar	Sablefish initial tag loss rate	Per tag
ϕ	$\in \mathbb{R}^+$	Scalar	Negative binomial dispersion	

mortality ($\mathbf{F}_t$ and $\mathbf{M}$ divided by 3 to reflect the quarterly time steps), respectively, η was stepwise ongoing tag loss, and λ was a calibration parameter to account for size-selectivity and any biases in fishing mortality. The stepwise fishing rates were constant across model steps within a given year and region.

We modeled expected tag recoveries by the matrix product of tagged sablefish counts and a step-wise observation rate. For a given release time-step and size class, this was defined as:

$$\hat{Q}_{i,d,l} = \mathbf{A}_{i,d,l} \mathbf{B}_{n[i,d],l}, \quad \text{Eq. 5}$$

where $\hat{Q}$ was a square matrix of expected tag recoveries with rows indicating release region and columns indicating recovery region, and $\mathbf{B}$ was a diagonal matrix of observation rates defined by the element-wise vector product:

Table 2

Movement model parameters and associated prior probability distributions (where applicable).

Parameter	Description	Prior [bounds]
Δ	Movement rate matrix (stepwise)	Diag N(0.98, 0.05)[0,1]
S	Survival rate matrix (stepwise)	n/a (function of fishing, natural and tag mortality)
f	Fishing mortality rate (stepwise)	From assessments, CV = 0.1
m	Natural mortality rate (stepwise)	N(0.1, 0.01)
λ	Calibration parameter to account for selectivity	No prior; [0,1]
η	Tag loss rate (including tag-induced mortality) (stepwise)	N(0.02, 0.001)
v	Initial tag loss rate (per-tag)	N(0.1, 0.01)
θ	Tag reporting rate (per-tag)	N(0.4, 0.04); N(0.5, 0.05); N(0.3, 0.03) for AK, BC, and CC, respectively
P	Annual movement rate	Product of stepwise movement rate
ϕ	Negative binomial dispersion parameter	N(1, 0.5)

$$\text{diag}[\mathbf{B}_{n,l}] = \theta(1 - \exp[-\lambda f_n]), \quad \text{Eq. 6}$$

where θ was a row vector of per-tag reporting rates. Tag reporting rates are only estimated for the three large-scale management regions.

For a given release size class, we defined annual sablefish movement rates by a matrix power of the corresponding stepwise movement rates:

$$P_l = \Delta_l^K, \quad \text{Eq. 7}$$

where P_l was the square matrix of annual movement rates (Δ_l is a square matrix of stepwise movement rates, as defined above for Eq. 3), and K was the number of time-steps per year (four in this case).

We used the NB2 (Hilbe, 2011) parameterization of the negative binomial distribution with probability mass function to link observed recoveries with those expected from the model:

$$\text{NB2}[Q_*|\hat{Q}_*, \phi] = \binom{Q_* + \phi - 1}{Q_*} \left(\frac{\hat{Q}_*}{\hat{Q}_* + \phi} \right)^{Q_*} \left(\frac{\phi}{\hat{Q}_* + \phi} \right)^\phi, \quad \text{Eq. 8}$$

where Q_* and $\hat{Q}_*$ were the observed and expected tag recovery counts corresponding to a given release region, time-step, and length class, and recovery region and time-step.

Existing estimates of fishing mortality and demographic rates were used as prior information when fitting the movement model. We used fully-selected fishing mortality for Chatham Strait in Alaska (Carlile et al., 2002), British Columbia (Department of Fisheries and Oceans, 2020) and the California Current (Haltuch et al., 2019). Fishing mortality in Alaska federal waters was estimated from catch and fully-selected (age 4+) biomass (described in Goethel et al., 2022). We used prior means of 0.1 for natural mortality and for initial tag loss (Hanselman et al., 2014) and 0.02 for ongoing tag loss including tag-induced mortality (Saunders et al., 1990); these assumptions were explored as a sensitivity. We set prior means for tag reporting rates (i.e., probability a tag is reported if recovered) to 0.4 in Alaska, 0.5 in British Columbia, and 0.3 in the California Current (Supplementary Figure S2); these were defined using the reporting rates from an exploratory data analysis of each region (rounding the average reporting rate across all years for each region to the nearest 0.1).

2.3. Model Fitting

The movement model was fit to the tagging data in a Bayesian estimation framework with Stan (Carpenter et al., 2017; “RStan, 2020), which implements the No-U-Turn Hamiltonian Markov chain Monte

Carlo algorithm (Hoffman and Gelman, 2014) for Bayesian statistical inference to generate the joint posterior probability distribution of all unknowns in the model. Values reported are posterior means with credible intervals unless otherwise stated. We sampled from three chains with 1000 iterations each and discarded the first quarter as warm-up. We assessed chain convergence visually via trace plots and by ensuring that the scale reduction factor $\hat{R}$ was less than 1.01 and that the effective sample size was greater than 400 (Vehtari et al., 2021).

2.4. Abundance Exchange

We used estimates of abundance (numbers by year) sourced externally from recent federal stock assessments (DFO, 2019; Goethel et al., 2022; Kapur et al., 2021) to convert movement rates estimated by our model to gross and net abundance exchange, as well as percentage of annual abundance, between Alaska, British Columbia, and the California Current on an annual basis. These abundance estimates excluded recruits (considered here as fish younger than age 2, consistent with the size classes present in the tagging studies which generally correspond to older ages). We combined state and federal estimates for Alaska. These comparisons are for illustrative purposes, given differences in assessment approaches across regions, though all regional assessments ignored movement (precluding ‘double-counting’ movement for this analysis) and involved methods that allowed the numerical characterization of sablefish abundance. Sablefish that moved between Alaska and the California Current were assumed to move through British Columbia. To account for uncertainty in both the abundance and movement estimates, we drew 1000 samples from a normal distribution with the mean and standard deviation of abundance and estimated movement rates for each region and year (Figure S2). These measures arose from internally-estimated asymptotic uncertainty measures from each stock assessment model and were on a similar scale across regions, with coefficients of variation ranging from 7 % in British Columbia to 16 % in the California Current. The steps to calculate estimated exchange (in numbers) were: 1) to generate a distribution of estimated abundance by drawing 1000 samples from an assumed normal distribution for each region and each year using the mean and standard deviation defined above; 2) to generate a distribution of movement rates by year and among regions by drawing 1000 samples using the mean and standard deviation of the movement rates from the 3-area base model, which included first- and second-degree movement (i.e. Alaska to British Columbia as well as Alaska to the California Current), and 3) calculating the product of these arrays for each unique year and sample (of 1000 samples). Net exchange was defined by the difference in absolute numerical transfers between regions in a given year. Therefore, the assumed abundance in a given region was explicitly adjusted by the movement rates to and from other regions individually, allowing us to express the numbers or percentage of abundance attributable to unique regional pairs. Uncertainty was calculated as the 5th and 95th quantiles for absolute or net movement in millions of fish each year and regional pair. The resultant array represented estimated exchange (in numbers) that accounted for uncertainty in both quantities, which could then be used to calculate the percentage abundance in each region and year that was attributable to movement from other regions.

2.5. Sensitivity Analyses

We performed several sensitivity analyses to evaluate the impact of model assumptions regarding size-specific movement, time-at-liberty, the ability for fish to transition between size classes, time blocks for fishing mortality rates, and the uncertainty of our parameter priors.

We selected these sensitivities for concision and clarity and therefore did not explore full combinations of changes to model structure. We provide three sensitivity analyses for the model described above, with a single size class. The first fit the original model structure to truncated

release time series which were split into three time blocks (1979–1994, 1995–2006, 2007–2018) to understand the extent to which movement rates vary across periods with very different fishery dynamics, and crudely over time. These time blocks correspond to the onset of the Individual Fishing Quota program in Alaska (starting in year 1995) and full integration of ITQs among fishing sectors accompanied by electronic monitoring in British Columbia (starting in year 2007), both of which would have impacted fishing dynamics. The second sensitivity analysis explored sensitivity to an increased uncertainty in fishing mortality (increasing the prior CV by a factor of three). The third sensitivity analysis separately explored sensitivity to greater uncertainty in the tag reporting rates by increasing the CV on each prior three-fold.

We then explored how movement estimates varied when the tag data were separated into release size classes into small (400–549 mm) and large (550 + mm) categories to coarsely represent immature and mature sablefish of both sexes across their range (Goethel et al., 2022). This definition does not fully capture the tendency for sablefish to mature at smaller sizes at more southerly portions of their range (Johnson et al., 2023). These size classes span 98 % of historical release sizes; we removed the 2 % of released data points that fell outside this range (Fig. 1). Estimates of movement rates in this fourth sensitivity model, which otherwise retained the assumptions of the base (pooled-sizes) model, therefore apply to the size of sablefish at the time of release, and not at the time of recovery. The fifth sensitivity analysis fit a three-area, two-size-class version of the movement model that did not constrain time at liberty so as to quantify sensitivity to our model assumption about time between release and recapture. The sixth and final set of analyses fit the two-size-class model to a dataset that excluded recoveries that had transitioned from the small to large size categories within three years in order to examine whether the confounding of growth and movement between tagging and recovery (size-class smearing) had a potential effect on estimated movement rates. This allowed for estimation of sized-based movement rates using one tag-recapture event per fish and per size class. Specifically, this meant an additional data-filtration step that removed individual fish that were below 550 mm at the time of release and above 550 mm at the time of recapture within the three-year time at liberty constraint. Altogether, these six sensitivity analyses incorporate key parameter and structural uncertainties for the base model, and allow for investigation of movement rate sensitivity to the consideration of sablefish size at release.

3. Results

Sablefish moved long distances for a groundfish species. The mean great circle distance between release and recovery, which is a conservative estimate of distance traveled, was 260 km between years one and two following release, and 346 km across all forty years of the study. Sablefish in the small length class moved 277 km between years one and two following release, and 215 km on average over all years. In contrast, sablefish in the large size category moved a mean distance of 246 between years one and two following release and 330 kms across all years. The furthest great circle distance measured was from a 520-mm fish (just shy of our threshold for the large size category) that was released in 1986 in the Aleutian Islands and recovered in the California Current approximately 4800 kms away, also in 1986. These findings indicate that most sablefish of small and large sizes can readily traverse international jurisdictions within one to three years (as the straight-line distance along the British Columbia Coast is roughly 950 km). Model convergence was achieved by all chains in our base and sensitivity models according to the ESS and visual inspection criteria mentioned above.

3.1. Movement Rates and Abundance Exchange in the Base Model

Annual retention rates were higher than transboundary movement rates for all regions (Fig. 2 and Table 3). The largest movement rates

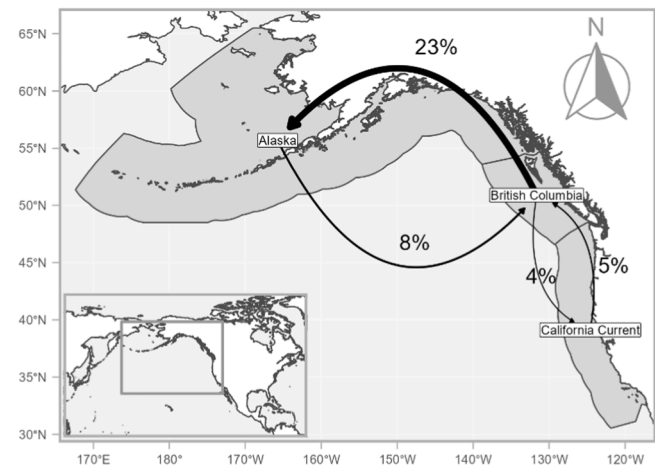


Fig. 2. Annual sablefish first-degree (between neighboring regions: black arrows) percentage movement rates (per fish per year) among three management regions in the northeast Pacific Ocean from the base model.

Table 3
Estimated annual movement rates (per fish for year; 90 % CI) for base model and size-based sensitivity model with two size classes between three management regions in the northeast Pacific Ocean. (AK: Alaska; BC: South British Columbia; CC: California Current). Movement is from row to column. Sensitivity run with three movement time blocks not shown here (but presented in Figures).

		Moving to		
		AK	BC	CC
Moving from	AK	Base Model (size invariant)		
		0.92 (0.91–0.93)	0.08 (0.07–0.09)	0.00 (0.00–0.00)
		0.23 (0.21–0.26)	0.73 (0.70–0.75)	0.04 (0.04–0.05)
		0.01 (0.00–0.01)	0.05 (0.04–0.06)	0.94 (0.93–0.95)
	BC	Sensitivity Model: Size-Structured, Small sablefish (<550 mm)		
		0.88 (0.75–0.96)	0.12 (0.04–0.24)	0.00 (0.00–0.01)
		0.23 (0.16–0.32)	0.72 (0.63–0.79)	0.05 (0.02–0.09)
		0.03 (0.02–0.04)	0.22 (0.17–0.29)	0.75 (0.68–0.81)
	CC	Sensitivity Model: Size-Structured, Large sablefish (≥550 mm)		
		0.69 (0.60–0.78)	0.28 (0.20–0.37)	0.02 (0.02–0.03)
		0.17 (0.12–0.24)	0.68 (0.60–0.75)	0.14 (0.10–0.19)
		0.02 (0.02–0.03)	0.24 (0.19–0.30)	0.73 (0.67–0.79)
	AK	Sensitivity Model: Size-Structured and no TAL limit, Small sablefish (<550 mm)		
		0.95 (0.94–0.95)	0.05 (0.05–0.06)	0.00 (0.00–0.00)
		0.10 (0.09–0.11)	0.89 (0.88–0.90)	0.01 (0.01–0.01)
		0.00 (0.00–0.00)	0.04 (0.04–0.05)	0.95 (0.95–0.96)
	BC	Sensitivity Model: Size-Structured and no TAL limit, Large sablefish (≥550 mm)		
		0.94 (0.94–0.95)	0.06 (0.05–0.06)	0.00 (0.00–0.00)
		0.12 (0.11–0.13)	0.85 (0.84–0.87)	0.02 (0.02–0.03)
		0.00 (0.00–0.00)	0.02 (0.02–0.02)	0.98 (0.97–0.98)
	CC	Sensitivity Model: Size-Structured and no transitions between classes, Small sablefish (<550 mm)		
		0.99 (0.99–0.99)	0.01 (0.01–0.01)	0.00 (0.00–0.00)
		0.31 (0.25–0.37)	0.68 (0.61–0.73)	0.02 (0.01–0.02)
		0.01 (0.01–0.01)	0.05 (0.04–0.07)	0.94 (0.92–0.96)
	AK	Sensitivity Model: Size-Structured and no transitions between classes, Large sablefish (≥550 mm)		
		0.91 (0.90–0.92)	0.09 (0.08–0.10)	0.00 (0.00–0.00)
		0.22 (0.19–0.24)	0.74 (0.71–0.76)	0.04 (0.04–0.05)
		0.00 (0.00–0.01)	0.04 (0.03–0.05)	0.95 (0.94–0.97)

were exhibited by British Columbia, with 27 % emigration (95 % CI 25 %–31 % emigration), most of which (23 %) was northward into Alaska. Movement rates between British Columbia and the California current were nearly identical (4 % and 5 %, 95 % CI 4 %–6 %), while 8 % of Alaskan sablefish were estimated to move southward into British Columbia (95 % CI 7 %–9 %). However, British Columbia was a net

recipient for abundance exchange: net exchange between Alaska and British Columbia was southward into British Columbia (averaging ~4.41 million fish per year, 95 % CI 1.96–6.95 million) while net abundance exchange between British Columbia and California Current was northward into British Columbia (averaging ~7.47 million fish per year, 95 % CI 4.75–10.40 million, Fig. 3). This corresponds to 20 %–40 % of annual abundance in British Columbia is attributable to immigration from each neighboring region individually. These findings are driven by differences in abundance among the three regions: despite showing the highest emigration rate, net abundance exchange was into British Columbia because of the higher numerical abundance of sablefish in Alaska and California Current. These findings suggest that British Columbia acts as a net recipient of sablefish abundance from both neighboring regions, and that this trend has been consistent throughout the available data period.

3.2. Sensitivity Analyses

We designed a suite of sensitivity analyses to investigate key structural assumptions of our model, and to characterize how consideration of sablefish size at tag release impacts uncertainty in movement estimates. Conducting three independent models at the time blocks defined by major changes in fishery dynamics resulted in movement rates that bracketed estimates from the base model for all management regions (Supplementary Figure S4). For Alaska, all time blocked movement models suggested similar or greater movement rates than the base model, which increased from 6 % in the first time block to 15 % in the latest time block (versus 8 % in the base model). Movement rates out of British Columbia were much higher for the earliest time block (34 % versus 27 % in the base model) and similar to the base model (26 % and

28 % respectively) in the latter time blocks (Figure S4). The California Current did not demonstrate a large change in movement rates for the first two time blocks (9 %–3 % for 1979–2006, versus 5 % in the base model), though the movement rate for the final time block (2007–2017) was higher (34 %) with higher emigration northwards (31 % versus 5 % in the base model) to British Columbia (Figure S4). Increasing the uncertainty on all fishing mortality or tag reporting rate priors three-fold in the three-area model resulted in similar movement rates for all regions, with only slightly higher residency rates for British Columbia (Figure S5).

The second set of sensitivity analyses investigated how movement estimates varied when tag releases were separated into immature and mature size classes, and when key structural assumptions that would affect size-based estimates were modified. For the sensitivity that separated tag releases by size class, movement rates and associated uncertainties were higher than in the base model for both size classes for all regions, with a most pronounced difference for the California Current: 25 %–28 % average movement versus 6 % movement for the base model that did not consider size class (Fig. 4). Large sablefish had greater movement rates than small sablefish for all regions, but the discrepancies among size classes in directional movement varied by region. Alaska had the highest retention rate for small sablefish (88 %) with only 12 % movement of small sablefish emigrating southward to British Columbia. More than double this percentage (28 %) of large sablefish were estimated to emigrate southward. British Columbia had the largest annual movement rates for both size classes, with 28 % of small sablefish and 32 % of large sablefish emigrating. Small sablefish were estimated to move north from British Columbia at approximately ~5-times the rate at which they move south to the California Current (23 % vs. 5 % per year, with a similar ratio for the pooled model,

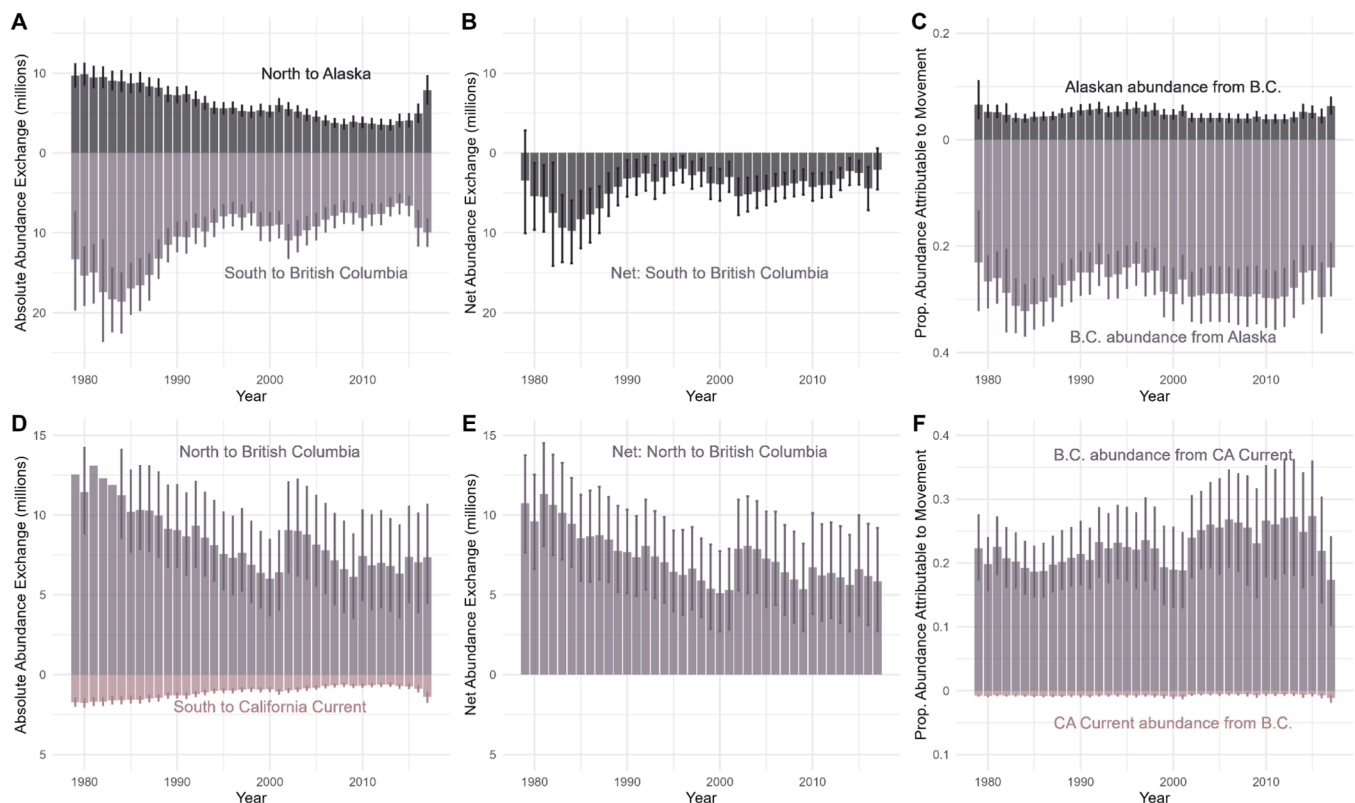


Fig. 3. A, B, D, E) Time series of absolute (A & D) and net (B & E) abundance exchange (90 % CI) in millions of sablefish between Alaska and British Columbia (top row, panels A & B) and between British Columbia and the California Current (bottom row, panels D & E). The rightmost column displays the proportion of regional abundance in a given year (90 % CI) attributable to movement between Alaska and British Columbia (top row, panel C) and British Columbia and the California Current (bottom row, panel F). Confidence intervals arise from resampling ($n = 1000$) the distribution of numerical abundance (from recent stock assessments) and movement rate posteriors from our base model.

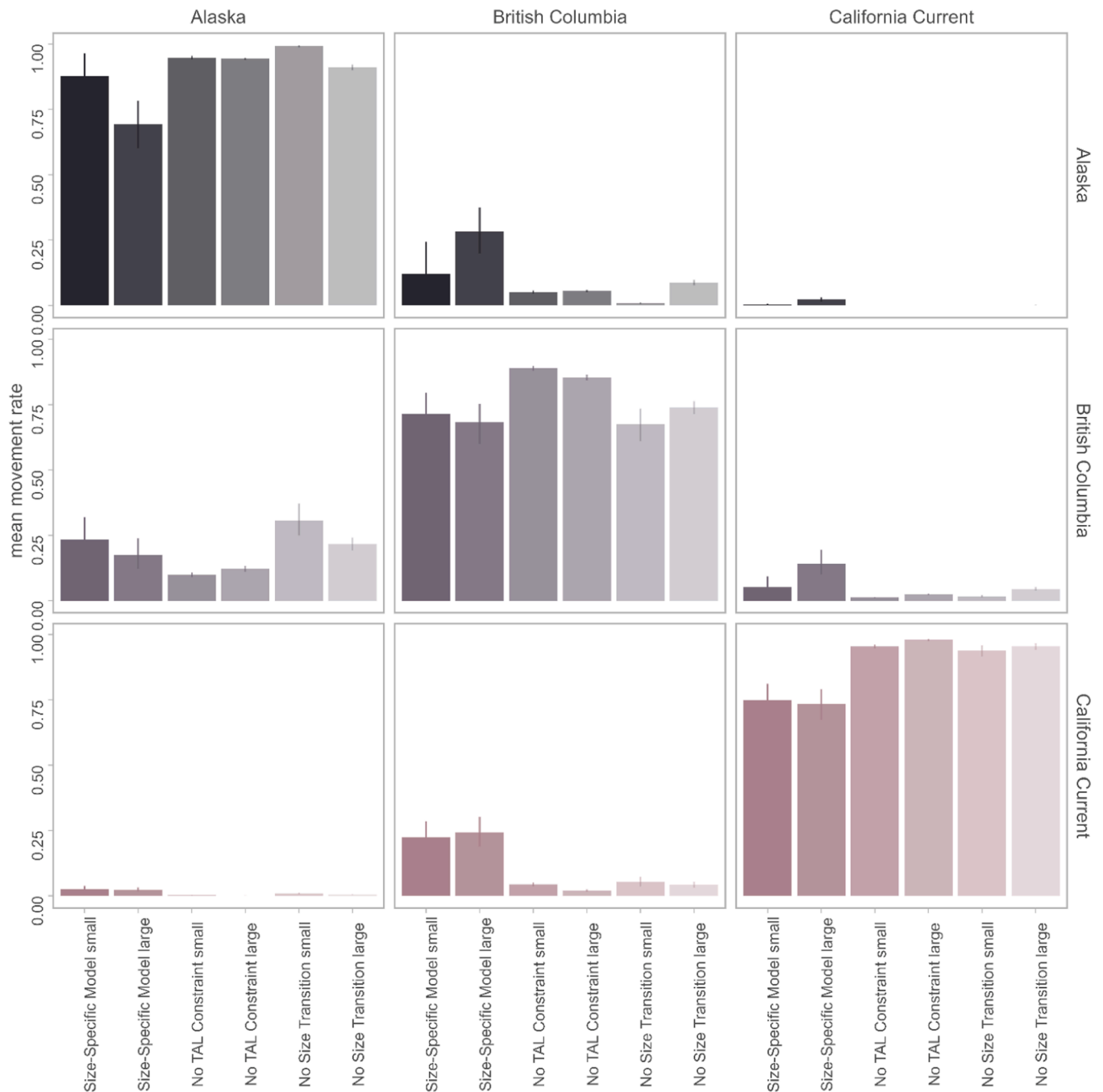


Fig. 4. Sensitivity analyses with two size classes. Sablefish mean annual movement rates (90 % CI) between three management regions in the northeast Pacific Ocean by released size class, using the same model structure as in the base model (“Size-specific Model”), removing the three year time at liberty constraint (“No TAL Constraint”) or removing recaptures that had transitioned between size classes (“No Size Transition”). Panel rows: previous (source) management region; panel columns: current (recipient) management region. Small: fork length 400–549 mm; Large: fork length 550–800 mm.

Table 3). Large sablefish did not exhibit such a strong discrepancy in movement rates (17 % northward and 14 % southward from British Columbia (Fig. 4 and Table 3). Movement northward from the California Current was consistent across size classes (22 % of small sablefish and 24 % of large sablefish moving northward into British Columbia.

Movement estimates from the two-size-class model appeared more similar to the one-size-class (base) model for both size classes when the three year time-at-liberty constraint was removed (Fig. 4) or tagging data for fish that transitioned between size classes were removed (Fig. 4). For these model runs, uncertainty in movement rate estimates were lower. Removing recoveries that transitioned between size classes accentuated movement rate differences by size class for British

Columbia, diminished size-class differences in Alaska, and was unchanged for the California Current (Fig. 4).

4. Discussion

Our study reinforces and extends prior knowledge about the high mobility and transboundary nature of northeast Pacific sablefish. Across four decades of conventional tagging data, we observed significant movement across management boundaries, with realized absolute exchange quantified through integration with stock assessment abundance estimates. Movement probabilities varied regionally and moderately by size class, providing new insight into broader-scale spatial population

dynamics and their implications for stock assessment and management.

Directional movement was evident across several region pairs. That work did not characterize net migration. While directional movement from British Columbia to Alaska was detected in our study, net demographic leakage was into British Columbia. This result was driven by abundance gradients: Alaska and the California Current host larger sablefish populations, and thus even modest movement rates from these regions result in meaningful biomass moving into British Columbia. This suggests that British Columbia operates as an important transition zone for sablefish migrating from south to north and that British Columbia receives a large biomass subsidy from these regions, particularly from Alaska. This result must be contextualized by the increased reporting rates in British Columbia beginning in 1997, when 100 % observer coverage was implemented (as reflected in the reporting rate priors for this model, [Figure S1](#)), which render estimates after that year more reliable than those before. Because we calculated abundance exchange in terms of numbers of fish, rather than the number of tags, this characterization is a straightforward result of the much larger abundances of sablefish present in the California Current and Alaska. This outcome persisted even when the range was divided into smaller spatial strata ([Figure S6](#)).

Overall, results from this study are largely consistent with previous research, where estimated movement rates were high and differed moderately across size, with larger individuals more likely to move compared to smaller ones ([Heifetz and Fujioka, \(1991\)](#); [Maloney and Sigler, \(2008\)](#); [Beamish and McFarlane, \(1988\)](#); [Hanselman et al., 2014](#); [Kimura and Shimada, 1998](#)). Our results suggest a general northwest-southeast movement pattern with age, whereby smaller fish tend to move towards Alaska and larger fish tend to move towards British Columbia and the California Current. The only other study to consider rangewide sablefish movement ([Kimura and Shimada, 1998](#)) suggested that movement rates were “probably sufficient to consider sablefish a single biological population...” but that independent sub-populations between Alaska and the U.S. West Coast could be characterized by demographic differences (namely size-at-age and -maturity), with British Columbia as a mixing area between them. That study also hypothesized that sablefish exhibit ontogenetic movement patterns based on their maturity levels, whereby immature fish move westward (for example, into Alaska from British Columbia, or across the Gulf of Alaska) and larger fish move eastward and southward as they mature. This hypothesis would correspond with our study’s observation that British Columbia receives a net positive input of abundance and retains a stable exploitable population through time, operating as a transition zone between regions.

Our sensitivity analyses suggest that, for the base model, increasing the uncertainty of the prior for fishing or reporting rates did not greatly influence movement rates or uncertainty thereof for any region ([Figure S3](#)). Adjusting the model structure to instead consider three time blocks only modified movement rate estimates for the California Current in the latest time period, for which there were very few releases ([Fig. 1](#)) and therefore much greater uncertainty in the movement estimates ([Figure S5](#)). For non-movement parameters (natural mortality and tag reporting rate), uncertainty did not vary considerably across sensitivity models (<5 % change in standard deviations); the only exception to this was for the sensitivity model for which the prior CV for tag reporting rates was increased by a factor of three, which had the expected impact of increasing uncertainty in this parameter estimate, but trivial impacts on movement rates. These findings suggest that the overall movement dynamics estimated by our base model are robust to uncertainties in key model specifications.

With respect to movement rates at size, we found that estimating movement rates for immature and mature size classes within the time-at-liberty constraints of the main model resulted in higher movement rates that were somewhat less certain than the base model for all regions. Again, this discrepancy was most pronounced for the California Current (which had less data overall, [Fig. 1](#)), and disappeared when sablefish

that transitioned between size classes were removed from the analysis ([Figure S5](#)). It is sensible that the base model’s movement estimates more closely match estimates for small sablefish, particularly in Alaska and British Columbia, given that they are more numerically abundant both in the population and the tagging data ([Fig. 1](#)). This suggests that average annual transboundary movement rates are being driven by immature sablefish. However, given that large (and therefore heavier) sablefish appear to move at higher rates than small sablefish, particularly from Alaska to British Columbia, it’s possible that future refinements of relative abundance exchange in terms of biomass would underscore the perception of British Columbia as a recipient of biomass from Alaskan sablefish. To summarize: large sablefish appear to move at higher rates than smaller sablefish, but because they are fewer and less represented in the tag data, the net effect of small sablefish movement on numerical abundance is greater. Future analyses could clarify the net effect in terms of biomass, which will likely highlight the importance of the movement of larger sablefish. The size-based sensitivities indicate that our base model is internally consistent and provides insight for future investigations.

Our findings have important implications for both sablefish stock assessment and management especially since all three regions are currently managed as separate populations ([Kapur et al., 2024](#)). These findings build upon previous, smaller scale investigations of sablefish movement in that they quantify transboundary migration for the first time, with explicit investigation of differences in movement rates by size (via our sensitivity analysis) and a quantification of net exchange across international boundaries. As demonstrated by [Berger et al. \(2021\)](#) unidirectional movement of individuals between areas of differing productivity has the potential to result in biased estimates of biomass in stock assessments, particularly if management boundaries do not align with the biological population (which may be the case for Northeast Pacific sablefish). Furthermore, if the British Columbia region moderately relies on immigration to sustain abundance, maintaining sablefish at sustainable levels in Alaska and California Current may be necessary for ensuring population stability and resilience across these regions ([Kerr et al., 2010](#)), as well as ensuring sustainable distribution of the resource across international boundaries. In a similar vein, distinguishing whether increases in regional productivity stem from synchronous recruitment events or demographic leakage from population-sources will be essential for determining sustainable harvest levels in each region.

4.1. Assumptions and Limitations

In earlier iterations of this model (results not shown), several variations of model design and prior specification were attempted and then abandoned due to problems with model convergence, stability, or data availability. One such iteration attempted to estimate fully time-varying movement (unique movement rates for each time step among all regions), but was precluded by temporal inconsistencies in data availability particularly from the California Current. Specifically, intermittent spatial closures in the southern part of that region led to high variability in tag data from that region, sometimes including zero releases, which likely contributed to the inability for even simple three-area movement models to achieve convergence. Future research to investigate changes in movement rates over time could be isolated to exchange between Alaska and BC, which was smaller than the range-wide scope of the present study. Our sensitivity analyses utilizing time blocks or increasing the uncertainty on movement rate priors led to more residence (less movement) between these regions suggesting that the current model structure may accentuate the movement rates between Alaska and British Columbia. Because fishing mortality and natural mortality are understandably confounded in such models, we chose to not run sensitivity analyses on both parameters as we would expect them to impact the model similarly. Importantly, the time blocks investigated were based on historical fishery dynamics, and given the

scale mismatch between regions (in terms of fishery removals, tag recoveries, and absolute abundance) it is possible that precise estimates of movement rates between these regions will remain sensitive to model assumptions. We chose to keep model priors broad and/or uninformative whenever possible: for example, the movement rate priors are centered near full retention as was done for earlier sablefish work (Hanselman et al., 2014) and in simulation studies (Bi et al., 2020), so the estimates presented here are driven by the data. We tuned non-movement parameter priors (such as the tag reporting rate) by simple exploratory data analyses, as specified above, after finding that uninformative priors led to poor convergence; uncertainty in these prior specifications were explored via sensitivity analyses. All sensitivity analyses suggested the same overall dynamic described above.

Another limitation to this study was the necessary aggregation of tagging data from multiple, independent programs. This introduces the possibility that recoveries in regions other than where tags were released may have been less likely to be reported at all, which would have suggested lower movement rates. Similarly, California Current data in the final years of this study were limited, which may have also precluded findings of inter-regional movement and resulted in lower exchange estimates.

4.2. Future Work

The current modeling approach utilized all known extant tagging data for Northeast Pacific sablefish in an attempt to provide the most robust estimates of movement within and among regional management and assessment units. Despite collating one of the richest and longest running mark-recapture data sets for a marine species known to the authors, the limitations of conventional tagging data impeded implementation of more complex movement models. As with any modeling application, increasing the number of partitions in the model led to decreased sample sizes for the estimation of parameters, which was further exacerbated by unbalanced sampling designs and the nature of fishery-dependent tag returns. Future work will aim to increase the resolution of inference from tagging data by utilizing electronic tag types (e.g., satellite tagging) in addition to continued deployment of traditional tags (Echave, Katy, 2016). Electronic tags can provide estimates of full migration paths and do not necessarily rely on fishery recaptures, which makes them an informative complement to the existing tagging database. Additionally, electronic tags can help identify potential spawning migrations (e.g., through the use of spawning event tags that release when fish spawn) aiding identification of where and when sablefish spawn. Thus, it may be possible to determine whether sablefish exhibit homing behavior or site fidelity to specific spawning grounds, knowledge of which is a critical uncertainty in the understanding of sablefish life history (Rodgveller et al., 2015). Leveraging the multiscale information content from multiple tag types (e.g., traditional mark-recapture, electronic tags, and even natural tags, otolith micro-chemistry, or approaches using genetic data such as close-kin mark recapture) would enable much higher resolution estimates of movement and distribution (Thorson et al., 2021).

Tagging analyses are important for understanding the ecology of a species and potential transboundary movement dynamics, but integrating tagging data into spatially explicit assessment models represents the next necessary step in understanding regional population dynamics. Tag-integrated assessments leverage the information content of traditional assessment data (e.g., fishery catch, survey abundance indices, and age or length composition data) with that from tagging data, which can better inform population scale (i.e., absolute abundance), mortality, and movement (Goethel et al., 2019). Thereby, improved estimates of regional biomass, demographic leakage, and linkages among populations (e.g., common trends in recruitment) can be attained. For instance, a tag-integrated model for Alaskan sablefish has been developed (Cheng et al., 2025), which leverages information on spatial dynamics from both tagging and compositional data (i.e., age and length),

and has provided unique insight into the ontogenetic connectivity across management regions in Alaska that differs from insights from analyses using only tagging data (e.g., Hanselman et al., 2014). However, the ability to implement more broad-scale integrated assessments will likely depend on leveraging multiple tag types, while being able to fit data at multiple resolutions. Scale-free approaches that are able to fit and estimate high resolution movement parameters (e.g., from electronic tagging data) and easily scale up to regional dynamics (e.g., matching assessment or management units) will be essential for more robust determinations of age and time-varying movement and demographic leakage across boundaries (e.g., Thorson et al., 2021).

4.3. Conclusions

Our results demonstrated the high movement potential and transboundary nature of Pacific sablefish, which generally supports findings from previous tagging analyses (e.g., Hanselman et al., 2014; Kimura and Shimada, 1998). Although sablefish are known to interact across the three primary regional management boundaries (i.e., Alaska, Canada, and United States west coast) and have been shown to be a panmictic population (i.e., Jasonowicz et al., 2016; Timm et al., 2024), the resource is managed in isolation within the three regions (Rooper et al., 2021). However, boundary areas between Alaska, Canada, and the Northern California Current demonstrate low residency and a high degree of mixing (Fig. 2 and Table 3). The realized impact of movement across international boundaries on local population abundance is likely to differ among regions (Fig. 2), given the difference in magnitude of assumed local stocks (e.g., the British Columbia stock is estimated to be of much smaller size than the Alaska stock). When non-negligible abundance exchange occurs among management regimes, it is generally recommended that movement should be addressed within the assessment-management framework (Kerr et al., 2017). Thus, the Pacific Sablefish Transboundary Assessment Team (PSTAT) conducted a spatially explicit management strategy evaluation (MSE) to address whether the current 'management in isolation' approach is robust to estimated levels of abundance exchange (Kapur et al., 2024) and found it to be satisfactory on the regional scale, though with important risks to sub-regional population units. Ideally, MSE can be used in the future to identify a minimally complex and maximally robust management procedure that can provide sustainable harvest recommendations across the entire Northeast Pacific, potentially without needing to implement complex spatial assessments (Goethel et al., 2023a,b).

Continued and ongoing efforts through the PSTAT and similar multinational working groups will be important for ensuring sustainable harvest of wide-ranging species, such as sablefish. Moreover, it will be important for countries sharing living marine resources to collaborate to enable adequate monitoring, especially at range extremes, as dynamic ecosystem shifts continue to cause species redistribution (Karp et al., 2019). Therefore, coordinated efforts among management bodies may be necessary to account for the transboundary nature of this species. Establishing collaborative monitoring programs and data-sharing agreements could help unify assessment approaches, ultimately supporting the development of more robust spatial harvest strategies that are better aligned with the species' movement dynamics and population structure (Bosley et al., 2019; Kapur et al., 2024). Given that the number of transboundary and straddling stocks will likely continue to increase in the near future (Palacios-Abrantes et al., 2022) increased international collaboration and innovative solutions (e.g., adaptive and dynamic ocean management) will be needed to prevent conflicts and maintain sustainable exploitation (Maxwell et al., 2015; Mendenhall et al., 2020).

CRedit authorship contribution statement

Brendan Connors: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Conceptualization. **Maia Kapur:** Writing – review & editing, Writing – original draft,

Visualization, Formal analysis, Conceptualization. **Aaron M. Berger:** Writing – review & editing. **Katy Echave:** Writing – review & editing, Data curation. **Melissa A. Haltuch:** Writing – review & editing, Funding acquisition, Conceptualization. **Kendra R. Holt:** Writing – review & editing, Data curation, Conceptualization. **Philina English:** Writing – review & editing. **Luke Rogers:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Daniel R. Goethel:** Writing – original draft, Methodology, Conceptualization. **Matthew L.H. Cheng:** Writing – original draft, Visualization. **Dana Hanselman:** Writing – original draft, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Luke Rogers reports financial support was provided by Fisheries and Oceans Canada. Maia Kapur reports financial support was provided by National Marine Fisheries Service. Maia S. Kapur joined the editorial board of Fisheries Research in March of 2025. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.fishres.2025.107498](https://doi.org/10.1016/j.fishres.2025.107498).

Data availability

The data that has been used is confidential.

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